

Kerui Wu

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Education

University of Massachusetts Amherst (UMass Amherst)

Sept. 2025 – Present

Ph.D. in Computer Science, GPA: 3.90/4.0

Advisor: Prof. Mingda Qiao

Rensselaer Polytechnic Institute (RPI)

Sept. 2021 – May 2025

B.S. in Computer Science, GPA: 3.76/4.0

Magna Cum Laude

Research Focus

- **Learning Systems** (Reinforcement Learning, Online Learning)
- **AI Agents** (Sequential Decision-Making, Multi-Agent Coordination, Resource Allocation)

Publications

(* Co-first authors; ($\alpha - \beta$) alphabetic order)

- **Adversarially Batched Online Learning**
 - ($\alpha - \beta$) Mingda Qiao, Kerui Wu
 - Under Submission
- **Compute Allocation for LLM-based Evolutionary Search: From Depth-Breadth to Bandits**
 - Kerui Wu^{*}, Sixue Xing^{*}, Haoyu He^{*}, Zhuo Yang, Haozheng Luo, Tianfan Fu, Aarth Nagarajan
 - EMNLP 2026 (Acceptance Rate: 15%) — **Oral**
- **ClinicalReTrial: Clinical Trial Redesign with Self-Evolving Agents**
 - Sixue Xing, Kerui Wu, Xuanye Xia, Meng Jiang, Jintai Chen, Tianfan Fu
 - EMNLP 2026, Findings (Acceptance Rate: 29%)
- **On the Adversarial Robustness of Graph Neural Networks with Graph Reduction**
 - Kerui Wu, Ka-Ho Chow, Wenqi Wei, Lei Yu
 - ESORICS 2025 (Acceptance Rate: 16%) — **Oral**
- **Large EEG-U-Transformer for Time-Step level Detection**
 - Kerui Wu, Ziyue Zhao, Bülent Yener
 - Technical Report, arXiv — **Winning Model of 2025 Seizure Detection Challenge**
- **On the Robustness of Graph Reduction Against GNN Backdoor Attacks**
 - Yuxuan Zhu, Michael Mandulak, Kerui Wu, George Slota, Yuseok Jeon, Ka-Ho Chow, Lei Yu
 - AISec@CCS 2024 (Acceptance Rate: 25%)

Awards

NSF NAIRR Pilot Start-Up Award, PI

2026

National Artificial Intelligence Research Resource (NAIRR)

CICS Scholarship

2025

Manning College of Information & Computer Sciences, UMass Amherst

Best Algorithm Award

2025

2025 Seizure Detection Challenge, International Conference on AI in Epilepsy

Certified in CyberSecurity (CC)

2023

ISC² (International Information System Security Certification Consortium)

Research Experience

University of Massachusetts Amherst (UMass Amherst)

Sept. 2025 – Present

Advisor: Mingda Qiao

Ph.D. Research Assistant

Robust Reinforcement Learning & Online Learning

- Characterize RLVR training dynamics under LLM-as-a-judge and reward models with systematic verification errors, modeling noise effects on the training trajectory.
- Benchmark **noise-corrected reinforcement learning** across reward-noise regimes and develop ROM, a robust RL algorithm that recovers **clean-reward GRPO performance** under all evaluated noisy settings.

- Establish theoretical foundations for **adversarially batched online learning**, where policies can be updated only under restricted, adversarial update schedules.
- Design **provably optimal learning algorithms**, including a generic **adaptive-doubling scheme** that requires no prior knowledge of update times while preserving optimal performance guarantees.

AI Agents and Adaptive Compute Allocation

- Systematically characterize compute allocation in **LLM-guided self-evolution**, uncovering a fitness–compute envelope and **capability-dependent depth–breadth trade-offs** across models and tasks.
- Develop BaSE, an adaptive **Multi-Armed Bandits**-based framework that dynamically allocates LLM calls from stagnating runs to promising trajectories, achieving 12.3% fitness improvement across five LLMs (e.g., **Qwen3** and **Llama**) over three tasks.
- Build ClinicalReTrial, a reasoning-driven **LLM multi-agent system** that casts clinical trial redesign as an iterative protocol optimization problem.
- Combine **retrieval-augmented generation (RAG)**, **hierarchical memory**, and **in-context learning** for continual agent self-refinement, achieving an 83.3% protocol-improvement rate and 5.7% success probability improvement.

Rensselaer Polytechnic Institute (RPI)

Advisor: Lei Yu, Bülent Yener

Jan. 2024 – May 2025

Undergraduate Researcher

Efficient Representation Learning for Long-Sequence EEG

- Introduce a time-step-level **sequence modeling formulation** for long-sequence EEG, eliminating redundant sliding-window inference and enabling **event-centric prediction**.
- Design a scalable **Conv-Transformer architecture** with multi-scale CNN encoders, residual convolutional blocks, and **self-attention temporal embeddings**.
- Achieve a **10× runtime speedup** and a 19.4% F1-score improvement, winning the **1st Place** in the 2025 Seizure Detection Challenge at International Conference on AI in Epilepsy.

Adversarial Robustness of Graph Neural Networks

- Empirically study the **robustness of Graph Neural Networks** against poisoning, evasion, and backdoor attacks.
- Analyze the security implications of graph reduction for **scalable GNNs**, characterizing how structural compression changes the impact of adversarial perturbations.

Selected Projects

China Construction Bank Achievement System

React, Go, MySQL, Azure

- Developed a production web platform for tracking employee performance, **deployed across 11 branches of China Construction Bank**.
- Implemented full-stack functionality, including data dashboards, analytics queries, and role-based management tools.
- Designed secure authentication and API infrastructure using **SHA256 hashing and JWT**, and deployed the system on **Microsoft Azure cloud**.

Submittly Open-Source Contribution

PHP, JavaScript, SQL, Python

- Contributed to Submittly, an open-source **course management** and **automated grading** system adopted by multiple universities.
- Implemented **full-stack features** including customizable pronoun settings across frontend UI, backend APIs (PHP), and relational database schemas (SQL). **5 pull requests were merged into the main branch of the repository**.

Technical Skills

Machine Learning: PyTorch, Torch-Geometric, Scikit-Learn, vLLM, LangChain, MCP

Programming: Python, C/C++, JavaScript, Go, PHP, SQL

Services

Program Committee Member: ICLR 2027